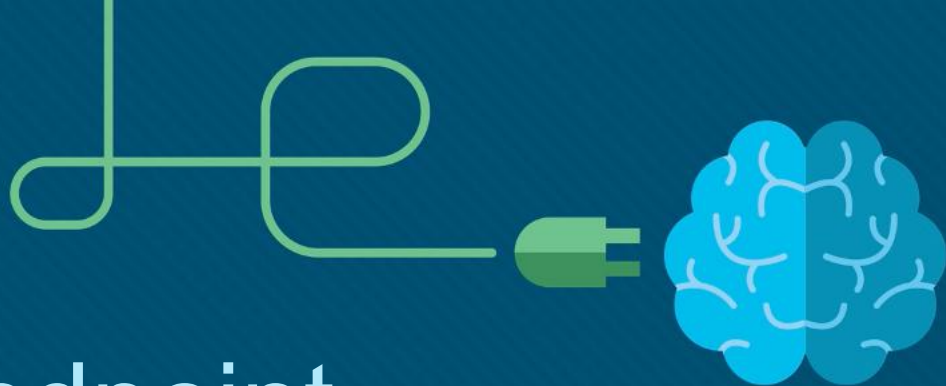




Module 13: Endpoint Security

Networking Security v1.0
(NETSEC)



Module Objectives

Module Title: Endpoint Security

Module Objective: Explain endpoint vulnerabilities and protection methods.

Topic Title	Topic Objective
Endpoint Security Overview	Describe endpoint security and the enabling technologies.
802.1X Authentication	Explain the functions of 802.1x components.

13.1 Endpoint Security Overview

Endpoint Security Overview

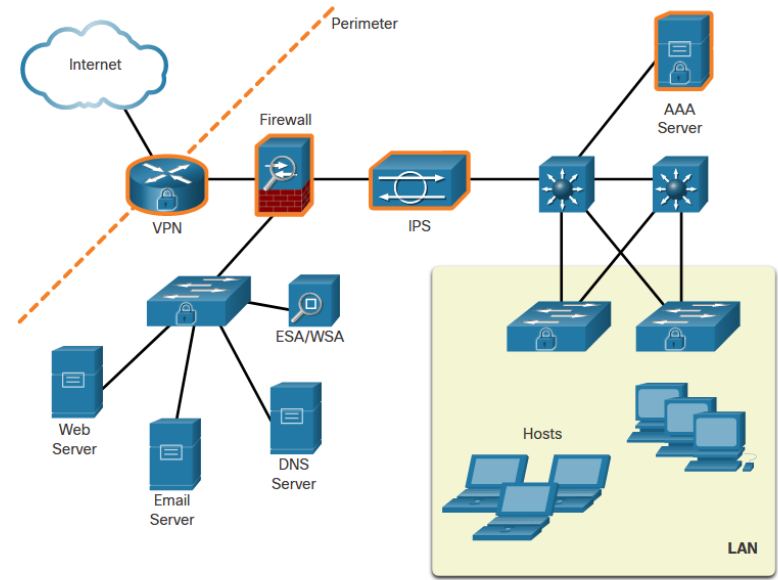
LAN Elements Security

Various network security devices are required to protect the network perimeter from outside access. As shown in the figure, these devices could include hardened ISR that is providing VPN services, an AS firewall appliance, an IPS, and a AAA server.

Because many attacks originate from inside the network, securing an internal LAN is just as important as securing the outside network perimeter

Specifically, there are two internal LAN elements to secure:

- Endpoints
- Network infrastructure



Traditional Endpoint Security

Historically, employee endpoints were company-issued computers which resided within a clearly defined LAN perimeter. These hosts were protected by firewalls and IPS devices which worked well with hosts that were connected to the LAN and behind the firewall.

The endpoints also used traditional host-based security measures:

- Antivirus/Antimalware Software
- Host-based IPS
- Host-based firewall

The Borderless Network

The new bring-your-own-device (BYOD) needs of workers have required a different way of approaching endpoint security.

In many networks, the network-based devices are disparate and typically do not share information among themselves. Additionally, new endpoint devices are not good candidates for the traditional host-based endpoint security solutions because of the variety of devices and the variety of operating systems available on those devices.

Security for Endpoints in the Borderless Network

Organizations must also protect their endpoints from new threats and provide the protection measures that are outlined in the table below:

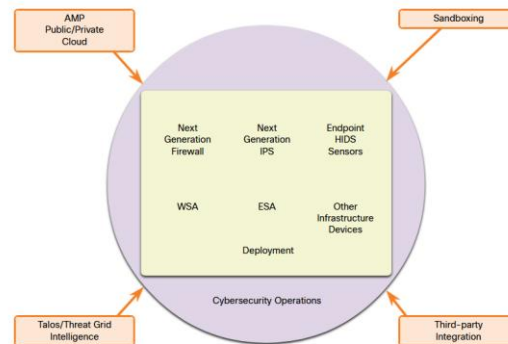
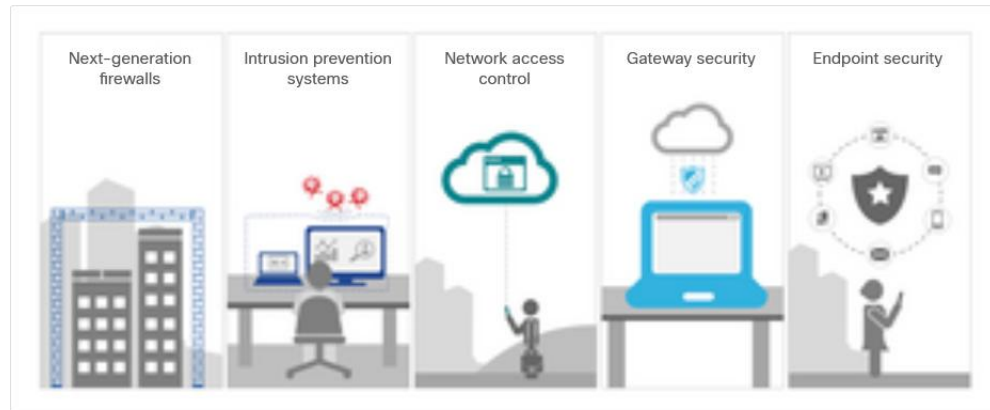
Measure	Purpose
antimalware software	Protect endpoints from malware.
spam filtering	Prevent spam emails from reaching endpoints.
blocklisting	Prevent endpoints from connecting to websites with bad reputations by immediately blocking connections based on the latest reputation intelligence.
data loss prevention (DLP)	Prevent sensitive information from being lost or stolen.

Network-Based Malware Protection

Protecting endpoints in a borderless network can be accomplished using network-based, as well as host-based techniques, as shown in the figure.

The following are examples of devices and techniques that implement host protections at the network level:

- Advanced Malware Protection (AMP)
- Email Security Appliance (ESA)
- Web Security Appliance (WSA)
- Network Admission Control (NAC)



Hardware and Software Encryption of Local Data

Endpoints are also susceptible to data theft. For instance, if a corporate laptop is lost or stolen, a thief could scour the hard drive for sensitive information, contact information, personal information, and more.

The solution is to locally encrypt the disk drive with a strong encryption algorithm such as 256-bit AES encryption. The encryption protects the confidential data from unauthorized access. The encrypted disk volumes can only be mounted for normal read/write access with the authorized password.

Network Access Control

The purpose of network access control (NAC) is to allow only authorized and compliant systems, whether managed or unmanaged, to access the network. It unifies endpoint security technologies with user or device authentication and network security policy enforcement.

NAC systems can have the following capabilities:

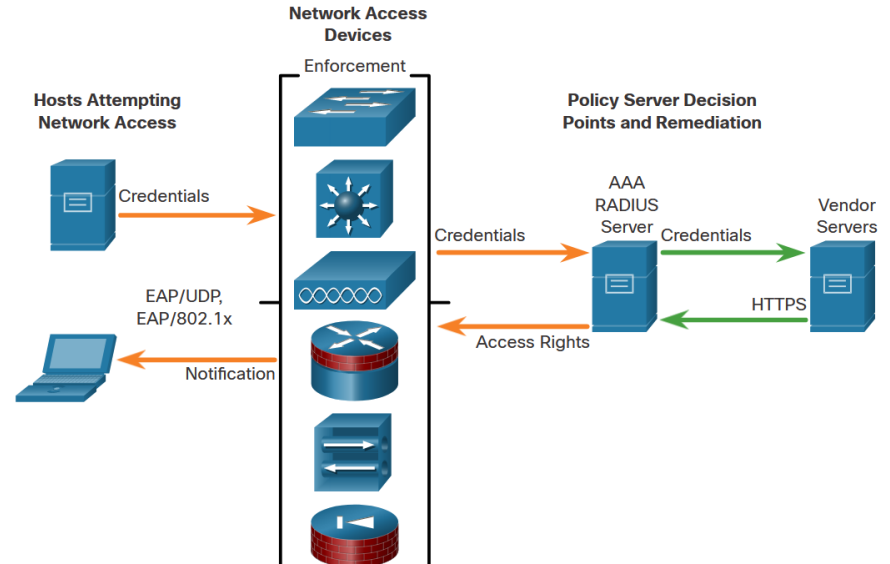
- **Profiling and visibility** – This recognizes and users and their devices before malicious code can cause damage.
- **Guest network access** – This manages guests through a customizable, self-service portal that includes guest registration, guest authentication, guest sponsoring, and a guest management portal.
- **Security posture checking** – This evaluates security-policy compliance by user type, device type, and operating system.
- **Incident response** - Mitigating network threats by enforcing security policies that block, isolate, and repair noncompliant machines without administrator attention.

Endpoint Security Overview

NAC Functions

The goal of NAC systems is to ensure that only hosts that are authenticated and have had their security posture examined and approved are permitted onto the network.

Network access devices can function as the enforcement layer, as shown in the figure. They force the clients to query a RADIUS server for authentication and authorization. The RADIUS server can query other devices, such as an antivirus server, and reply to the network enforcers.

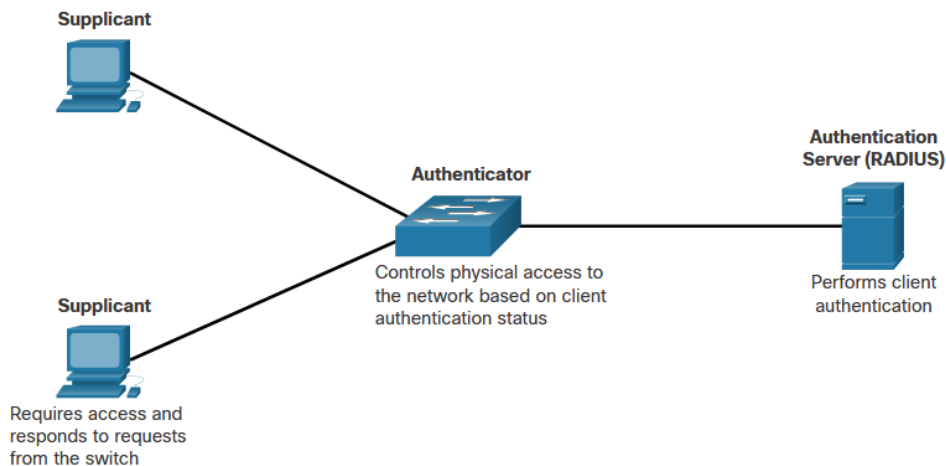


13.2 802.1X Authentication

Security Using 802.1X Port-Based Authentication

The IEEE 802.1X standard defines a port-based access control and authentication protocol that restricts unauthorized workstations from connecting to a LAN through publicly accessible switch ports. The authentication server authenticates each workstation that is connected to a switch port before making available any services offered by the switch or the LAN.

The figure shows that with 802.1X port-based authentication, the devices in the network have specific roles.



Security Using 802.1X Port-Based Authentication (Cont.)

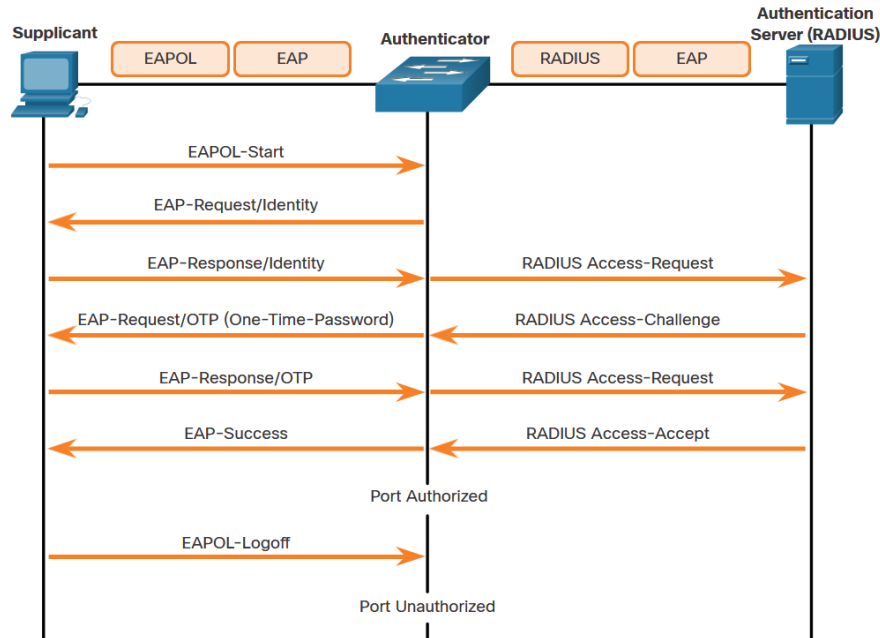
The 802.1x roles include:

- **Supplicant (Client)** - The device (workstation) that requests access to LAN and switch services and then responds to requests from the switch. The workstation must be running 802.1X-compliant client software. (The port that the client is attached to is the supplicant [client] in the IEEE 802.1X specification.)
- **Authenticator (Switch)** - Controls physical access to the network based on the authentication status of the client. The switch acts as an intermediary (proxy) between the client (supplicant) and the authentication server, requesting identifying information from the client, verifying that information with the authentication server, and relaying a response to the client. The switch uses a RADIUS software agent, which is responsible for encapsulating and de-encapsulating the EAP (Extensible Authentication Protocol) frames and interacting with the authentication server.
- **Authentication server** - Performs the actual authentication of the client. The authentication server validates the identity of the client and notifies the switch whether the client is authorized to access the LAN and switch services. Because the switch acts as the proxy, the authentication service is transparent to the client. The RADIUS security system with EAP extensions is the only supported authentication server.

Security Using 802.1X Port-Based Authentication (Cont.)

The figure shows the complete message exchange between the supplicant, authenticator, and the authentication server. The encapsulation occurs as follows:

- **Between the supplicant and the authenticator** - EAP data is encapsulated in EAPOL frames.
- **Between the authenticator and the authentication server** - EAP data is encapsulated using RADIUS.



802.1X Authentication

Control the 802.1X Authorization State

It may be necessary to configure a switch port to override the 802.1X authentication process. To do this, use the **authentication port-control** interface configuration command to control the port authorization state. The syntax for the command and a description of the parameters are shown below.



Parameter	Description
auto	Enables 802.1X port-based authentication and causes the port to begin in the unauthorized state. During this time only EAPOL, STP, and CDP frames are the only type of frames that can be sent or received through the port until the client device has been authenticated.
force-authorized	The port sends and receives normal traffic without 802.1x-based authentication of the client. This is the default setting.
force-unauthorized	Causes the port to remain in the unauthorized state, ignoring all attempts by the client to authenticate. The switch cannot provide authentication services to the client through the port.

802.1X Authentication

802.1X Configuration

802.1X Configuration requires a few basic steps:

Step 1. Enable AAA using the **aaa new-model** command.

Step 2. Designate the RADIUS server and configure its address and ports.

Step 3. Create an 802.1X port-based authentication method list using the **aaa authentication dot1x** command.

```
S1(config)# aaa new-model
S1(config)# radius server NETSEC
S1(config-radius-server)# address ipv4 10.1.1.50 auth-port 1812 acct-port 1813
S1(config-radius-server)# key RADIUS-Pa55w0rd
S1(config-radius-server)# exit
S1(config)#
S1(config)# aaa authentication dot1x default group radius
S1(config)# dot1x system-auth-control
S1(config)#
S1(config)# interface F0/1
S1(config-if)# description Access Port
S1(config-if)# switchport mode access
S1(config-if)# authentication port-control auto
S1(config-if)# dot1x pae authenticator
```

802.1X Configuration (Cont.)

Step 4. Globally enable 802.1X port-based authentication using the **dot1x system-auth-control** command.

Step 5. Enable port-based authentication on the interface using the **authentication port-control auto** command.

Step 6. Enable 802.1X authentication on the interface using the **dot1x pae** command. The **authenticator** options sets the Port Access Entity (PAE) type so the interface acts only as an authenticator and will not respond to any messages meant for a supplicant.

```
S1(config)# aaa new-model
S1(config)# radius server NETSEC
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S1(config-if)# switchport mode access
S1(config-if)# authentication port-control auto
S1(config-if)# dot1x pae authenticator
```

13.3 Endpoint Security Summary

What Did I Learn in this Module?

- Many devices and technologies enhance host-based endpoint protections including email security appliances, web security appliances, NAC, and the Cisco Identity Services Engine (ISE).
- Another way that endpoints can be protect from data loss is through the use of encryption using software, such as BitLocker.
- Network Access Control is a system that can check whether endpoints that attempt to the network comply with network security policies.
- Cisco ISE combines AAA and NAC and into a single system.
- 802.1X provides a means by which authenticator network access switch can act as an intermediary between a client and an authentication server.
- The system uses EAP and EAPOL to carry authentication traffic between the switch and the authenticator switch.
- 802.1X port-based authentication is configured by first globally activating AAA and by specifying the RADIUS server name, address, and ports. After that the authenticator interface is configured with 802.1X parameters.

New Terms and Commands

- | | |
|--|---|
| <ul style="list-style-type: none">• Antivirus/Antimalware Software• Host-based firewall• spam filtering• blocklisting• Advanced Malware Protection (AMP)• Email Security Appliance (ESA)• Web Security Appliance (WSA)• Network Admission Control (NAC)• security posture checking• incident response• 802.1X• supplicant• authenticator• authentication server | <ul style="list-style-type: none">• Extensible Authentication Protocol (EAP)• Extensible Authentication Protocol over LAN (EAPOL)• authentication port-control {auto force-authorized force-unauthorized}• aaa authentication dot1x default group radius• dot1x system-auth-control• authentication port-control auto• dot1x pae authenticator |
|--|---|

